On Analyzing Self-Driving Networks: *A **Systems Thinking** Approach*

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On Analyzing Self-Driving Networks: A Systems Thinking Approach

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Session Outline

1 The Problems: Motivating Systems Thinking

- *Complex Adaptive Systems*
- *Tame Problems vs. Wicked Problems.*
- *Problems Faced Due to Conventional Non-System Thinking*

2 The Solution: What Is Systems Thinking?

- *Defining Characteristics of Systems Thinking*
- *System Thinking Concepts of Leverage and Archetype*
- *System Thinking Tools*

3 Systems Thinking For The Internet and Future AI-Driven SDNs

- *The Paradoxes of Internet Design*
- *Ethical and Policy Challenges in SDNs.*

1

Motivating Systems Thinking

You think that because you understand "one" that you must therefore understand "two" because one and one make two.

BUT you forget that you must also understand "and."

So, what is a system?

A system must consist of three kinds of things:
elements, interconnections, and
a function or purpose."

A system is a set of things—**people, cells, molecules, or whatever**—interconnected in such a way that they produce their own pattern of behavior over time.

"The behavior of a system cannot be known just by knowing the elements of which the system is made."
— **Donella H. Meadows**, **Thinking in Systems: A Primer**

Complex Systems

Most important systems of interest are **complex** systems, and have the following properties:

Tightly Coupled

"Everything influences everything else" "You can't just do one thing"

Dynamic

Change occurs at many time scales

Policy Resistant

Many obvious solutions to problems fail or actually worsen the situation.

Counterintuitive

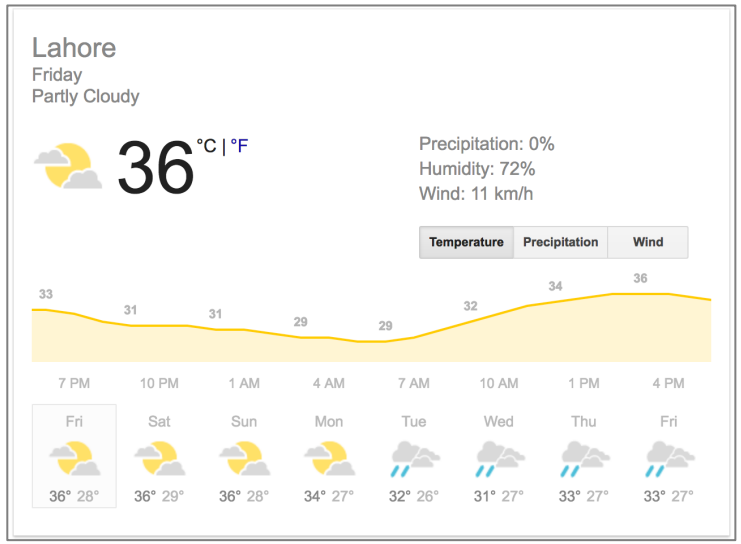
Cause and effect are distant in time and space

Exhibit Tradeoffs

Long term behavior is often different from short term behavior

Complex systems vs. Complex *adaptive* systems

weather prediction



In complex systems (e.g., weather), the system does not react to predictions about it.

stock market prediction



In complex adaptive systems (e.g., markets), the system reacts to predictions about it.

***Social systems are** typically nonlinear multi-loop feedback **complex adaptive systems**, which makes their behavior counterintuitive*

Tame Problems vs. Wicked Problems

Stubborn problems that refuse to go away and often become worse despite efforts and best intentions

Examples:

Racism, Poverty, Terrorism, Inequality, Water Management.

Why are these problems so stubborn?

Problems Faced Due to Conventional (Non-System) Thinking

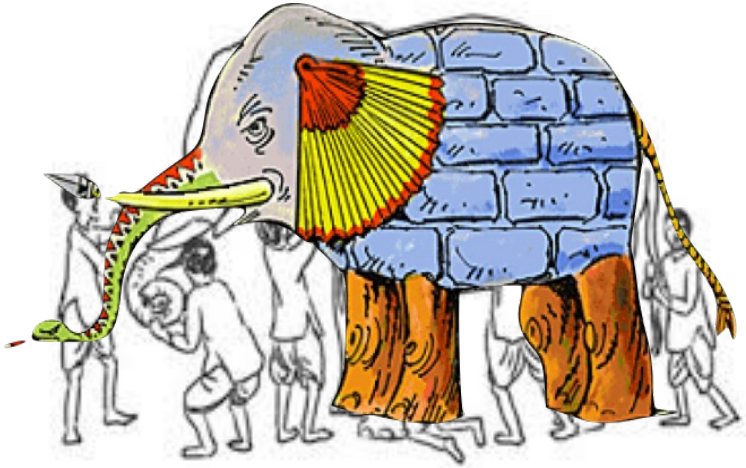
1. *Incorrect Mental Models;*
2. *Reductionism;*
3. *Policy Resistance;*
4. *Problem Solving Through Symptom Treatment*
5. *Unintended Consequences;*
6. *Optimizing the Parts at the Cost of The Whole;*
7. *Intervening and Causing Problems*

Mental models

A mental model is how we perceive reality. It comes through our backgrounds, values, expectations, and experiences.



Six blind men and the elephant



Incorrect assumptions

1. Linear Instead of Nonlinear;
2. Causal Links Rather Than Loops;
3. Immediate Feedback Rather Than Delayed Feedback;
4. Reductionism vs. Holism

Reductionism

A common way of doing analysis is to break down the system into parts and then analyze the parts.

But this approach suffers from a problem, which may be called the Humpty Dumpty Problem or the Broken Mirror Problem.

Once you've broken down the mirror to pieces, you cannot see the whole again.



HUMPTY-DUMPTY
sat on a wall,
HUMPTY-DUMPTY
had a great fall;
All the king's horses,
and all the
king's men
Cannot put Humpty-
Dumpty
together
again.



Policy Resistance

measures taken to improve a situation can directly make it worse due to policy resistance arising from people adapting

The Cobra Effect

(policy resistance and unintended consequences)



*Or the perils of **giving lollipops***

During the British Raj, a **bounty system** was devised to counter the rise of venomous cobras.

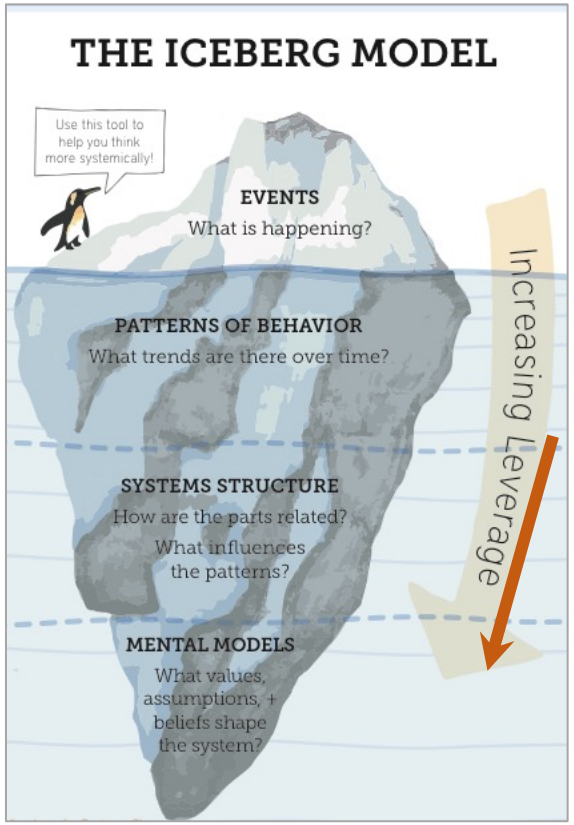
The system worked really well; a lot of cobras were killed.

Except that, entrepreneurs figured out they could make money by farming cobras and killing more of them.

After the government scrapped the system, ***there were more cobras than before*.**

Problem Solving Through Symptom Treatment

"There's always an easy solution to every problem that is neat, and plausible, ***but wrong****."—H. L. Mencken*

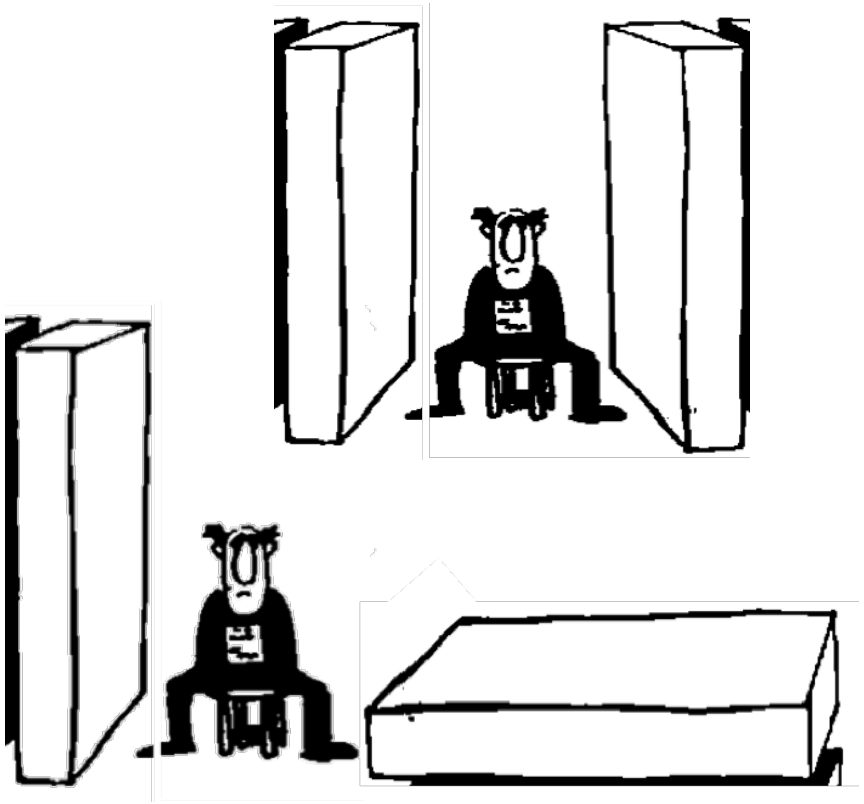


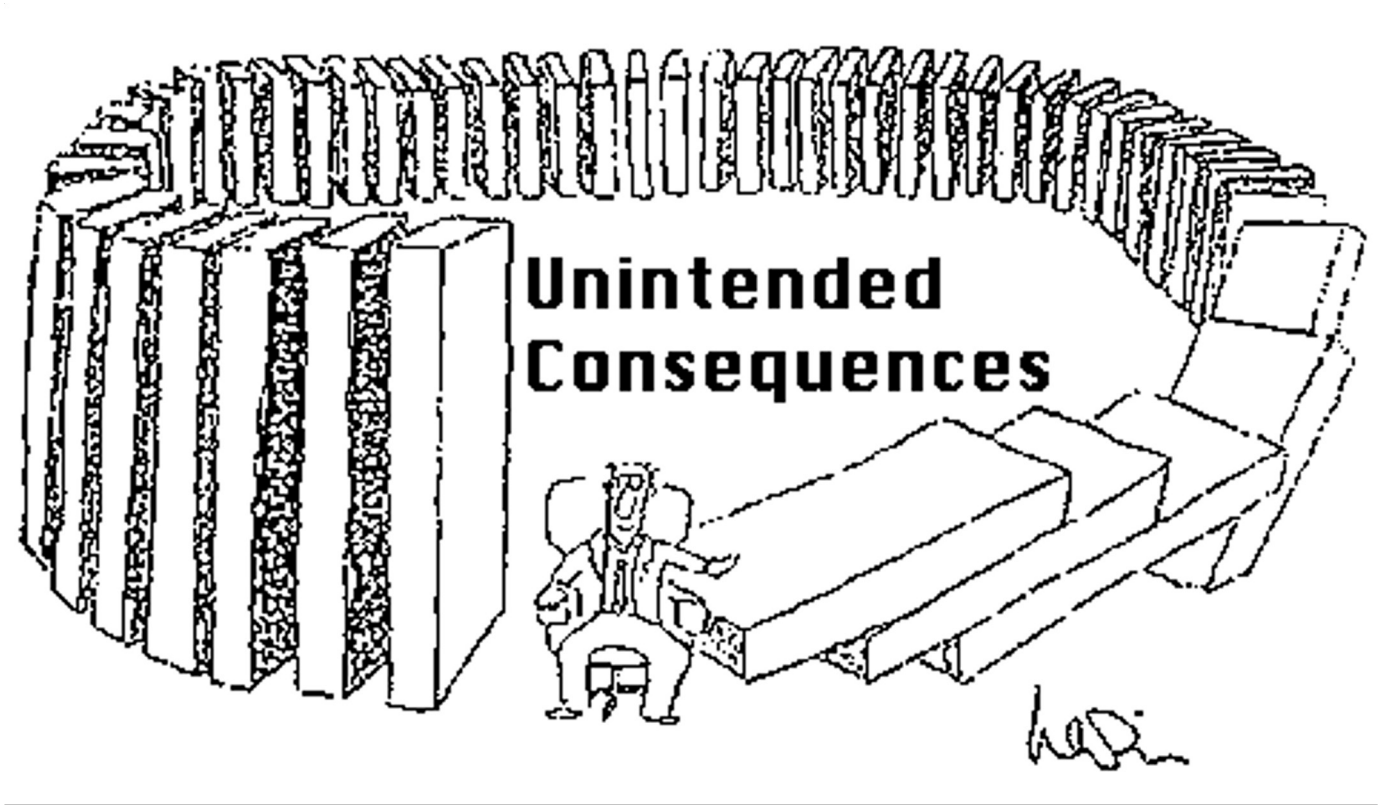
Leverage points are places within a complex system where a small shift in one thing can produce big changes in everything.

Interventions that address the systemic structure are more high-leverage than those that only address symptoms

Unintended Consequences

"There are downsides to everything;
there are unintended consequences to everything."—Steve Jobs

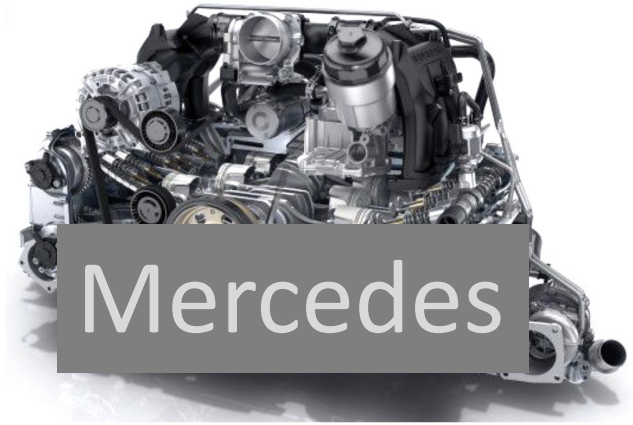




Optimizing the parts rather than the whole



+



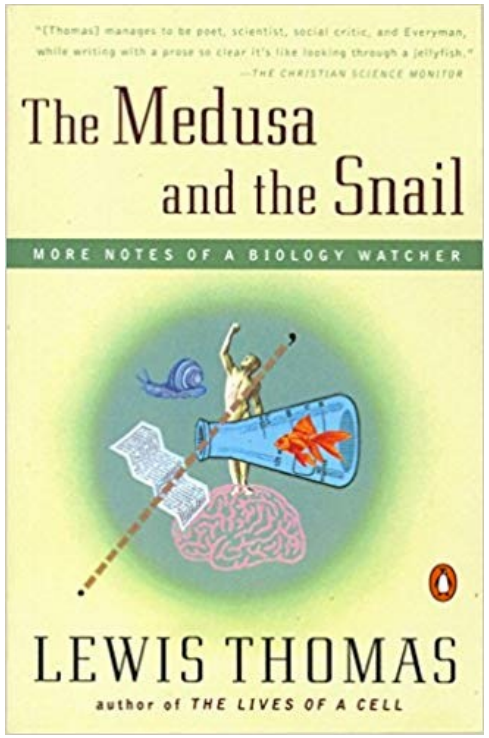


A star studded team can fail spectacularly if the parts don't fit well—or are not focused on the overall purpose.



Intervening and causing problems

"The chief source of problems is solutions"—Sevareid



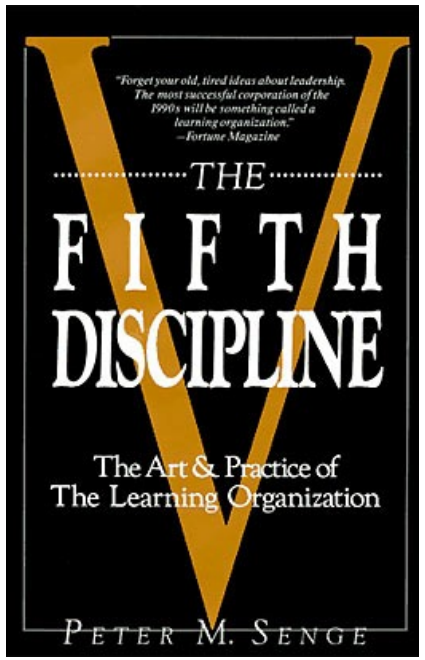
"When you are confronted by any complex social system, … you cannot just step in and set about fixing with much hope of helping. **This realization is one of the sore discouragements of our century** …

You cannot meddle with one part of a complex system from the outside without *the almost certain risk of setting off disastrous events that you hadn't counted on in other, remote parts.* If you want to fix something you are first obliged to understand … the whole system …

Intervening is a way of causing trouble."

In medicine, there's a special name for problems created by doctor's interventions. The term ***Iatrogenesis*** *refers to the "wounds brought forth by the healer."*

Peter Senge's laws of systems



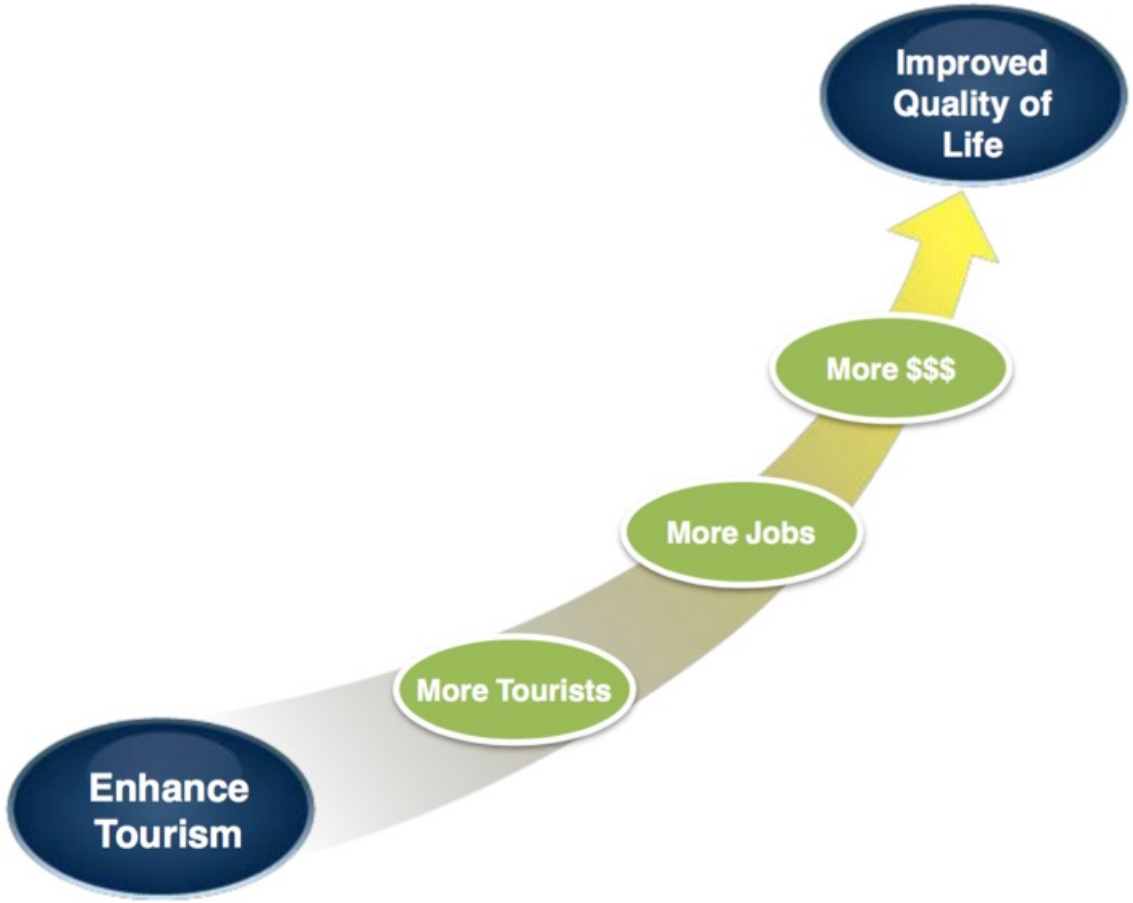
1) Today's problems come from yesterday's solutions.
2) **The harder you push, the harder the system pushes back.**
3) Behavior grows better before it grows worse.
4) **The easy way out usually leads back in.**
5) **The cure can be worse than the disease.**
6) Faster is slower.
7) **Cause and effect are not closely related in time and space.**
8) **Small changes can produce big results – but the areas of highest leverage are often the least obvious.**
9) You can have your cake and eat it too – but not at once.
10) Dividing an elephant in half does not produce two small elephants.
11) **There is no blame.**

2

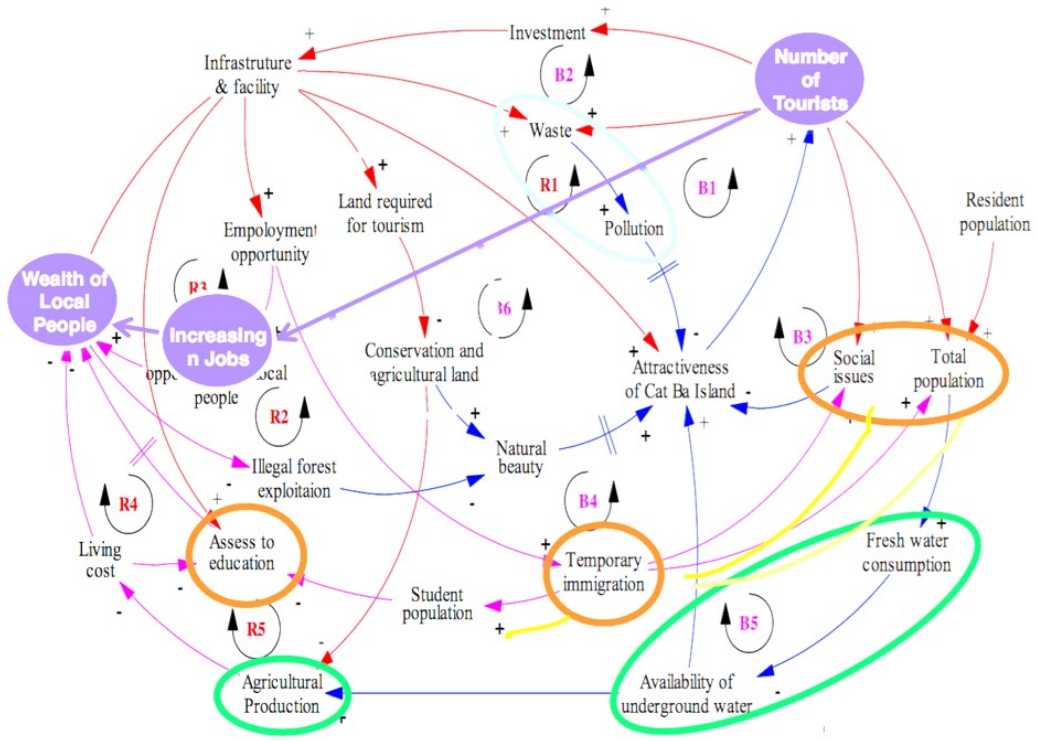
What Is Systems Thinking?

"Systems Thinking is the art and science of linking structure to performance, and performance to structure—often for purposes of changing structure (relationships) so as to improve performance"—Richmond.

Linear thinking *vs.* Systems thinking



*If we enhance tourism in a place, it will attract temporary immigrants which causes the population to increase creating its own problems (**social issues**, pollution, more resources needed, more crime, etc.).*



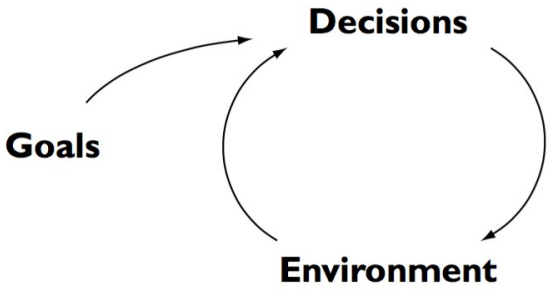
Feedback view of the world

"Everything we do as individuals, as an industry, or as a society is done in the context of an information-feedback system."—MIT's Jay Forrester

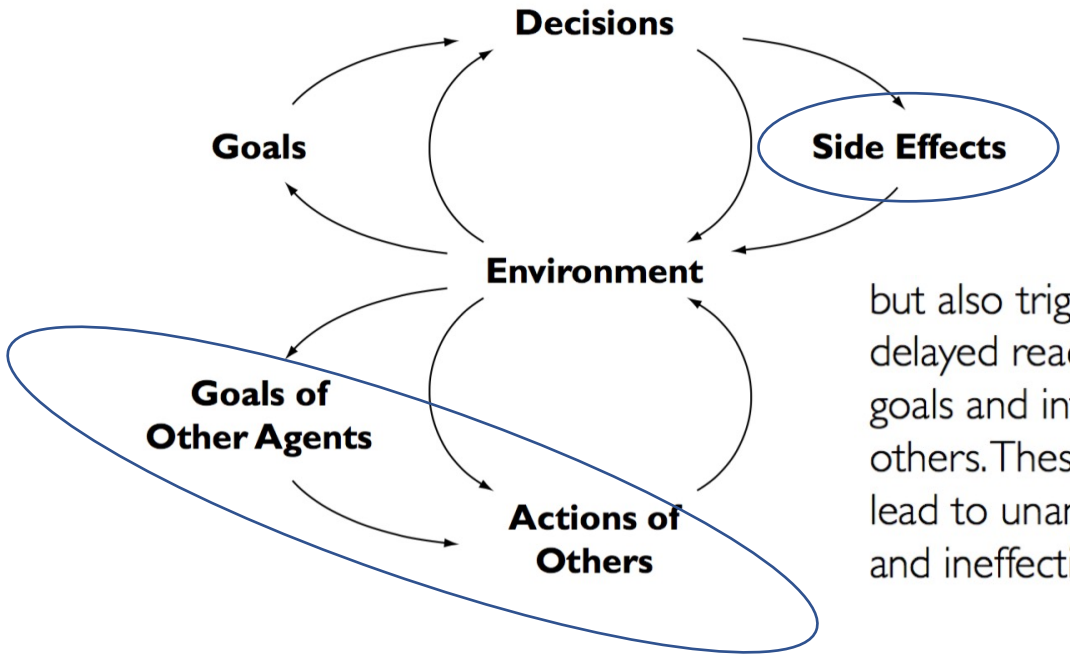
*The non-feedback view of the world (**Open-loop thinking**)*



Closed-loop thinking



Our decisions alter our environment, leading to new decisions,

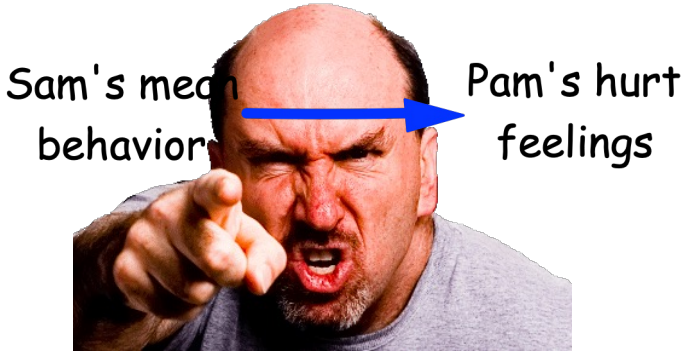


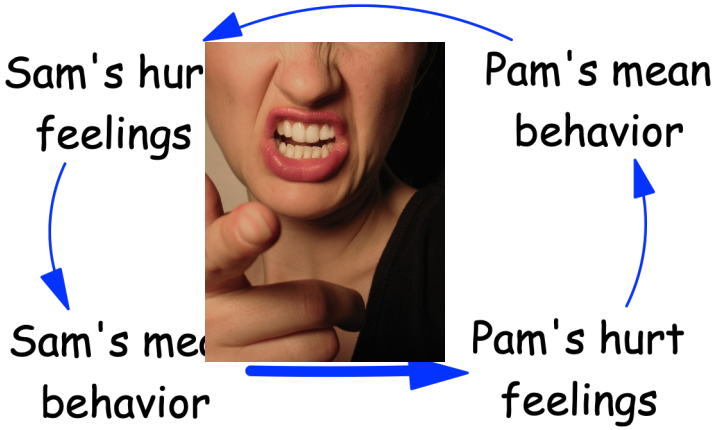
but also triggering side effects, delayed reactions, changes in goals and interventions by others. These feedbacks may lead to unanticipated results and ineffective policies.

System-as-a-cause thinking

"Systems are perfectly designed to achieve the results they are currently achieving." —Edward Deming







Exogenous point of view

Sam is always mean to Pam.

It's all his fault.

If he would be nicer, Pam's life would be better.

Endogenous point of view

Maybe there is something Pam is doing, which is causing Sam to be mean…

Exogenous: originating externally; ***Endogenous:*** originating internally

There's no blame! System-as-a-cause thinking

"Stop looking for who's to blame; instead you'll start asking, "What's the system?" The concept of feedback opens up the idea that a system can cause its own behavior." —Donella H. Meadows

The endogenous way is more empathetic and proactive



A famous quote in management:

"If you are not aware of how you are part of the problem, you can't be part of the solution."—Bill Torbert

Taking delayed feedback into account

We have the intuition that cause and effect are located in close in time and space. This is not always true in complex systems.

"Because of feedback delays within complex systems, by the time a problem becomes apparent it may be unnecessarily difficult to solve." — *Donella H. Meadows, Thinking in Systems: A Primer*







The Parable of the Boiled Frog



System thinking concepts

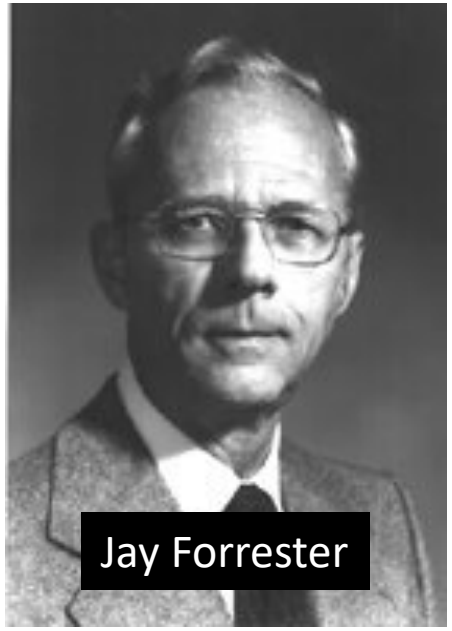
Leverage



Places within a complex system where a small shift can produce big changes in everything else

Concept of leverage

"Small changes can produce big results – but the areas of highest leverage are often the least obvious."—Peter Senge


Jay Forrester

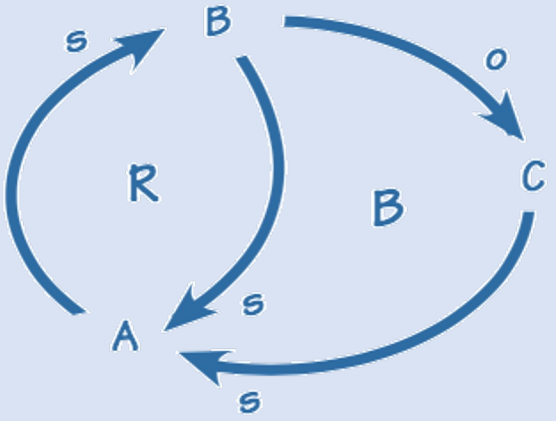
Pioneering research by Forrester showed that in many social systems, the high-leverage interventions were non-obvious and even if determined, liable to be pushed in the long direction

For example: to fix urbanization problems, we need to adjust low-cost public housing

*(we need **less** rather than more low-cost public housing)*

System Thinking Tools

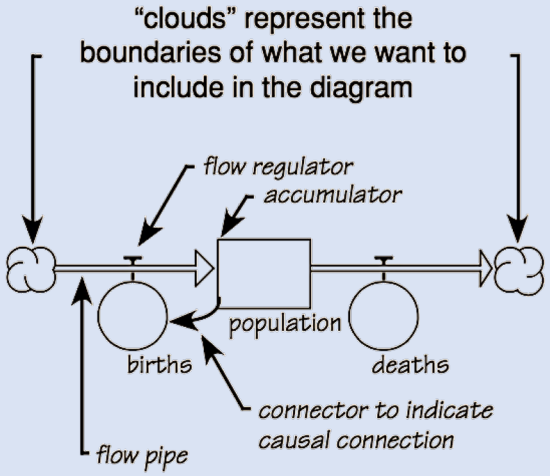
Qualitative tools



Causal loop diagrams

a framework for seeing interrelationships rather than things; can help in identifying reinforcing (R) and balancing (B) processes.

Quantitative tools



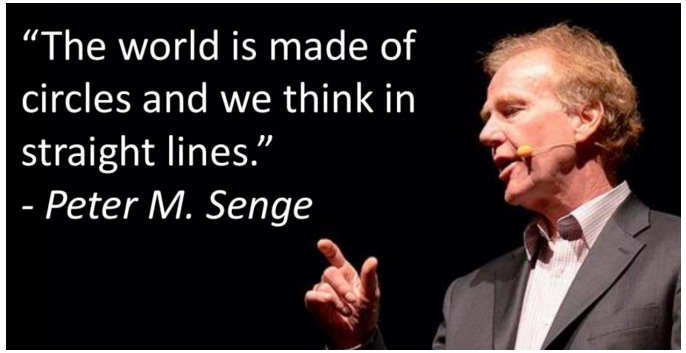
Stock and flow diagrams

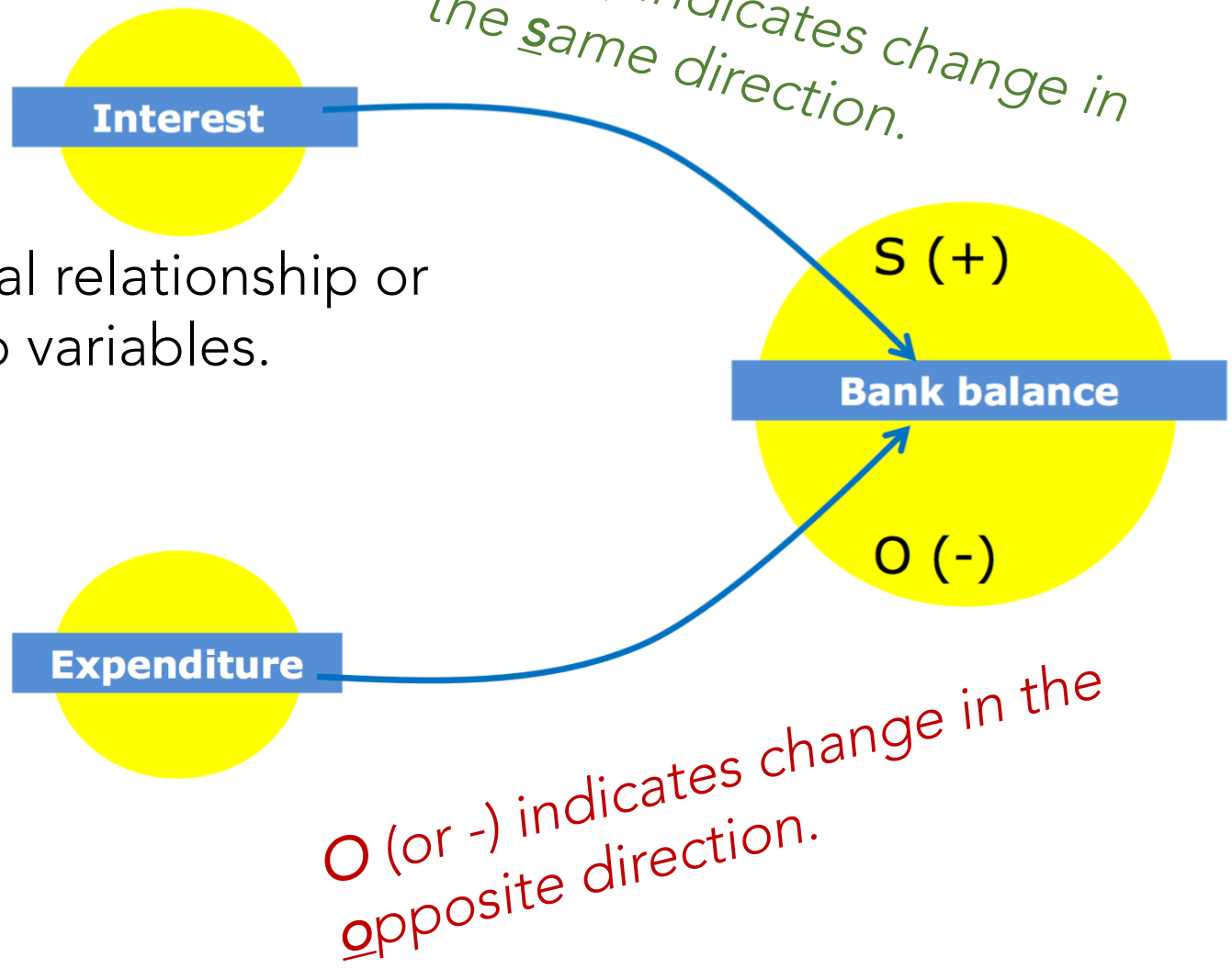
Unlike causal loops, stock and flow diagrams provide information about rates of change and accumulations.

System dynamics is grounded in control theory and the modern theory of nonlinear dynamics and offers many other rigorous tools

Causal links

An arrow indicates a causal relationship or change in the state of two variables.





Only two kinds of feedback loops

Reinforcing Loop (R)

Positive feedback

The R-loop represents growing or declining actions

Balancing Loop (B)

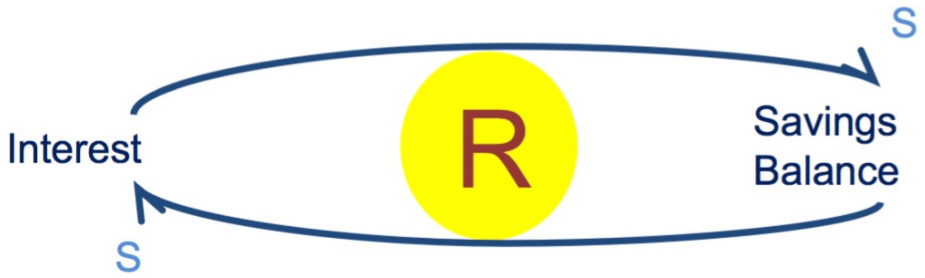
Negative feedback (counteracting)

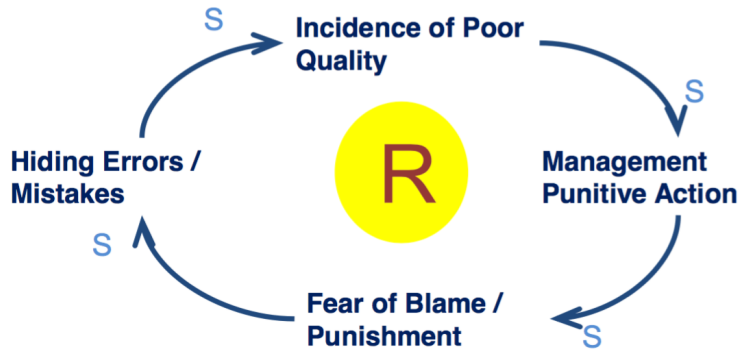
The B-loop seeks stability or return to control, or aims for a specific target

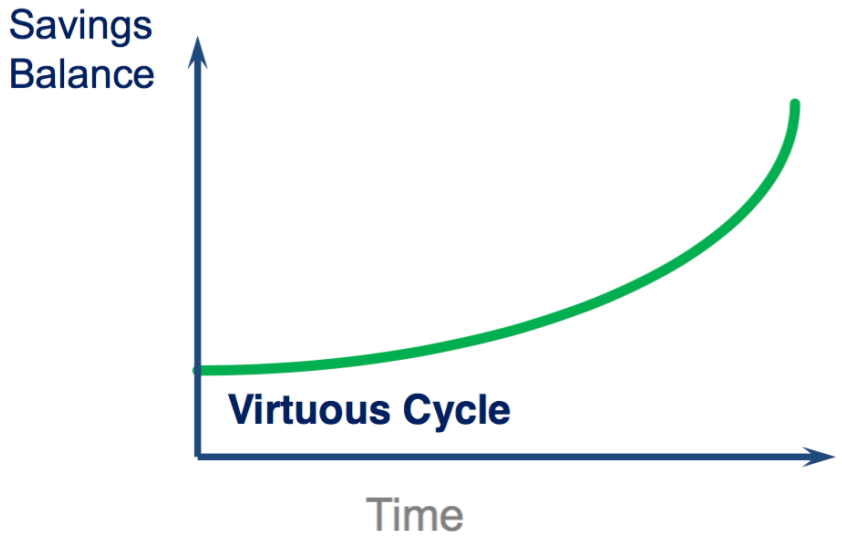
Reinforcing Loop (R)

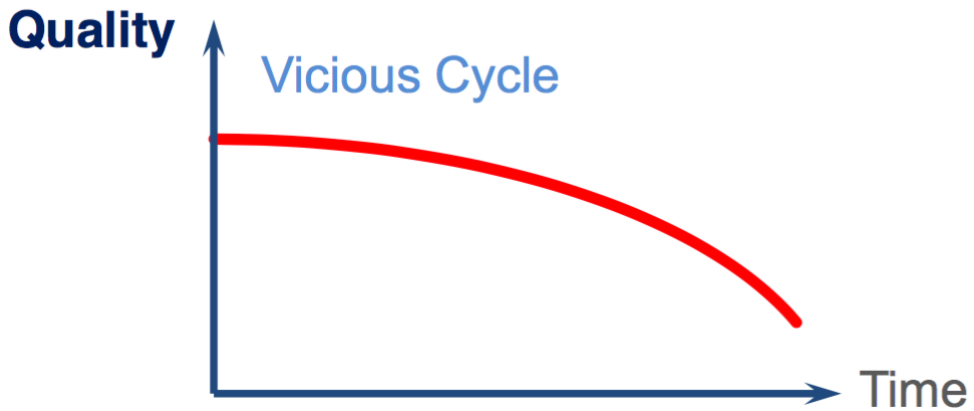
Positive feedback

The R-loop represents growing or declining actions





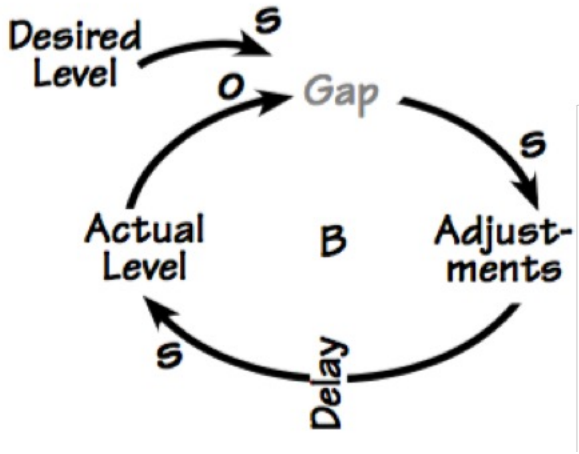


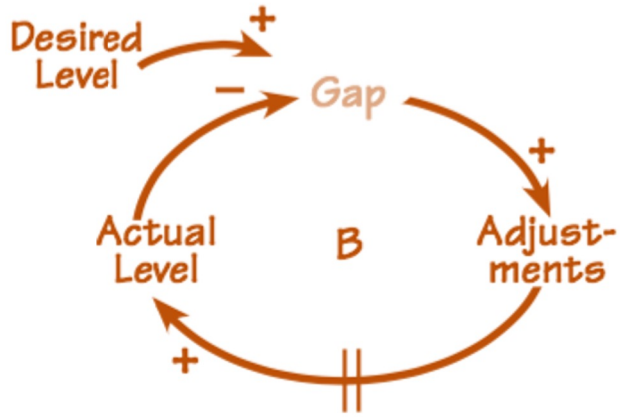


Balancing Loop (B)

Negative feedback (counteracting)

The B-loop seeks stability or return to control, or aims for a specific target



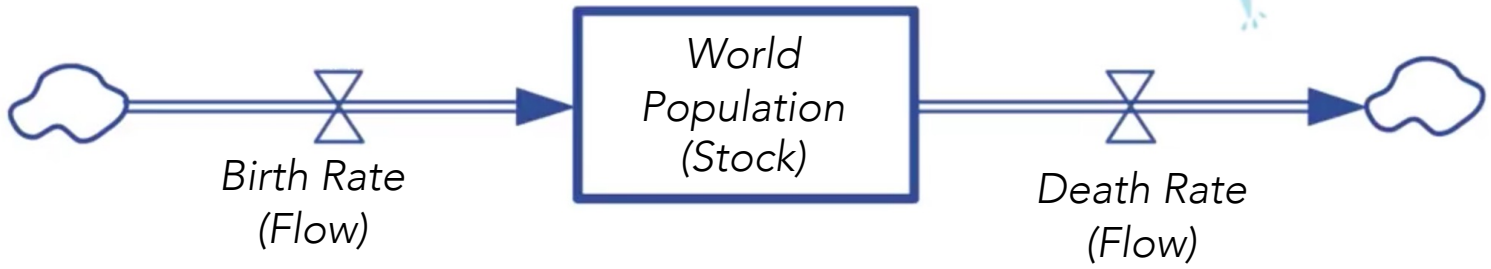


If there is a gap between the desired level and the actual level, adjustments are made until the actual equals the desired level. The starting variable is grey.

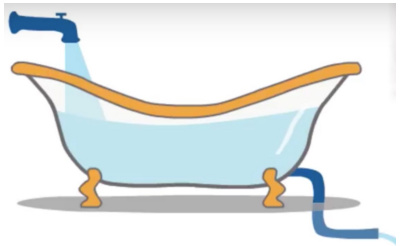
Thinking of stocks and flows

If the birth rates decline all over the world, will we continue to have increase in population growth?

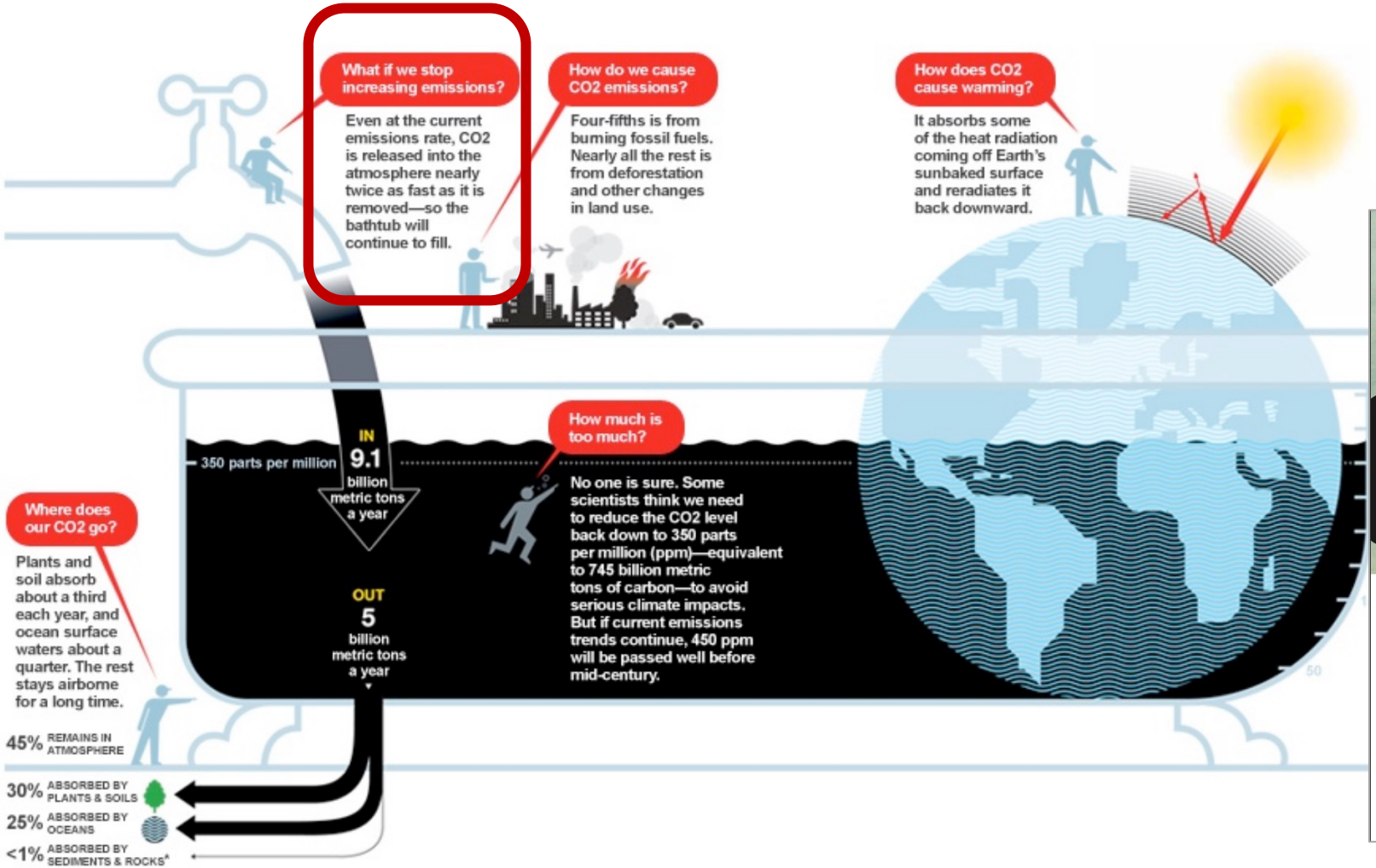
We can understand this using accumulations (or stocks) and rates (or flows)



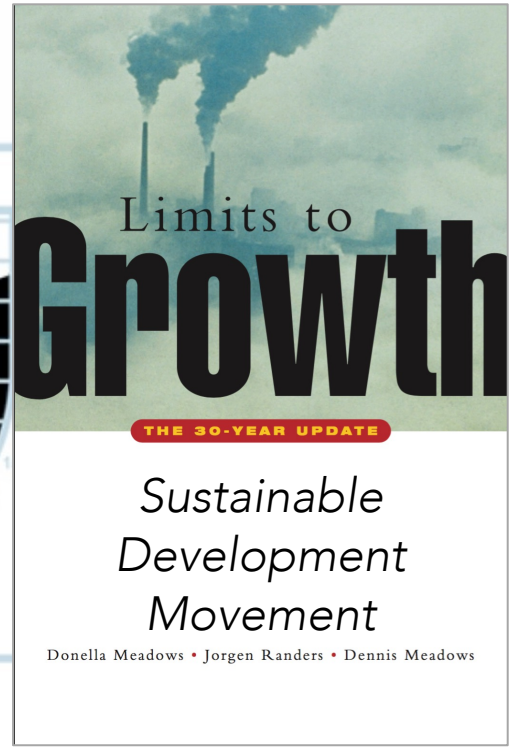
We can use the bath tub analogy to understand why despite global reduction in birth rates, the world population continues to increase.



The bathtub analogy was popularized by Sterman to clear the common misconception that CO_2 in the atmosphere will drop simply by reducing our emissions.



NATIONAL GEOGRAPHIC



System thinking concepts

Archetypes

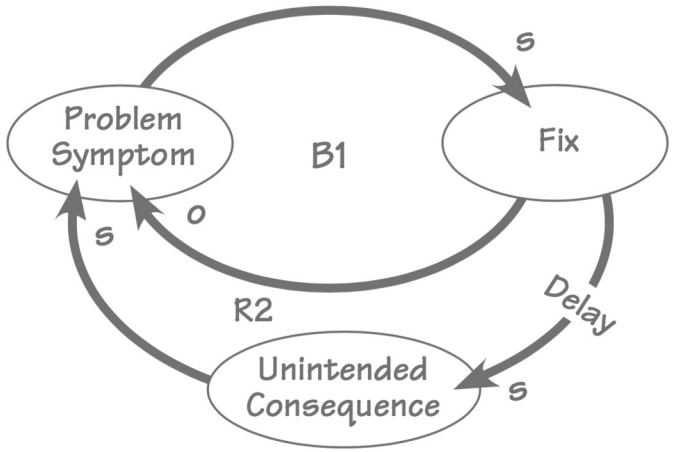


Archetypes can be used as templates for diagnosing vexing long-term problems.

1

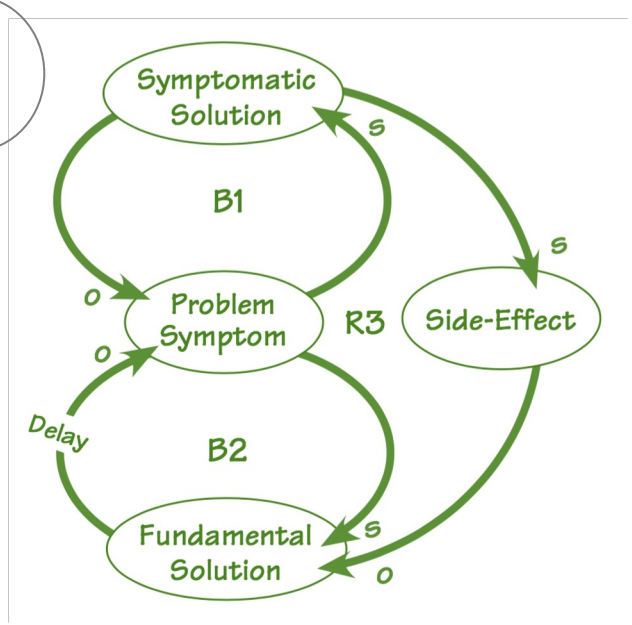
Fixes That Backfire

A quick solution with unexpected long-term consequences



In a **"Fixes That Fail" situation**, a problem symptom cries out for resolution. A solution is quickly implemented that alleviates the symptom (B1), **but the unintended consequences of the "fix" exacerbate the problem (R2).** Over time, the problem symptom returns to its previous level or becomes worse.

2



Shifting the Burden

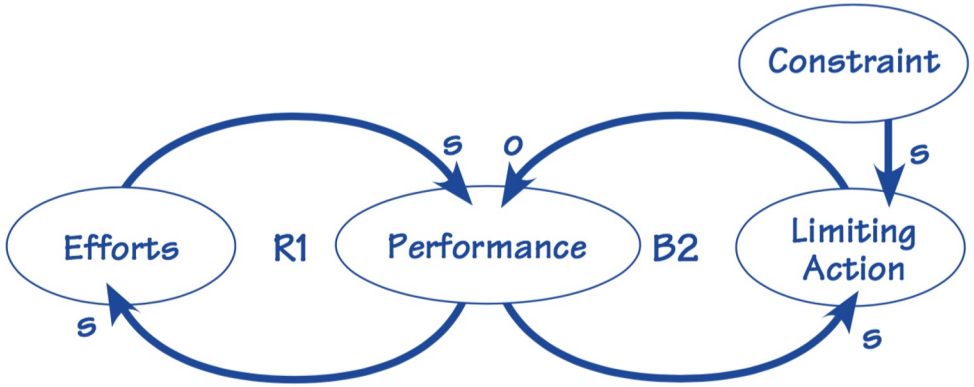
Systems unconsciously favor short-term, addictive solutions

In a **"Shifting the Burden,"** a problem is "solved" by applying a symptomatic solution (B1), which diverts attention away from more fundamental solutions (R3). In an "Addiction" structure, a "Shifting the Burden" degrades into an addictive pattern in which the side-effect gets so entrenched that it overwhelms the original problem symptom.

3

Limits to Growth

Improvement accelerates and then suddenly stalls

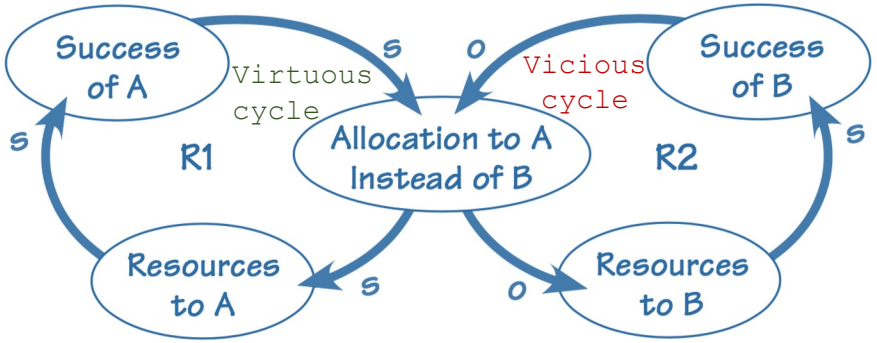


In a **"Limits of Success" scenario**, continued efforts initially lead to improved performance. Over time, however, the system encounters a limit which causes the performance to slow down or even decline (B2), even as efforts continue to rise.

4

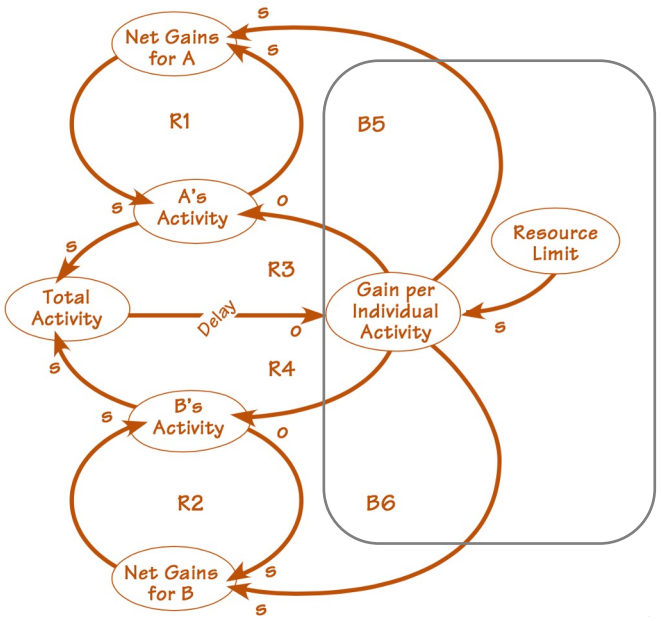
Success to the Successful

Things get better for "winners" and worse for "losers"



In a **"Success to the Successful"** archetype, if one person or group (A) is given more resources, it has a higher likelihood of succeeding than B (assuming they are equally capable). The initial success justifies devoting more resources to A, and B's success diminishes, further justifying more resource allocations to A (R2).

5

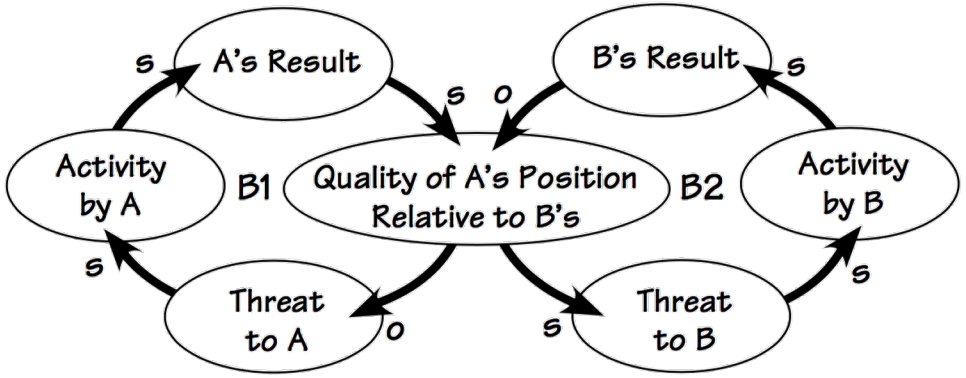


Tragedy of the Commons

Shared unmanaged resource collapses due to overconsumption

In a **"Tragedy of the Commons"** structure, each person pursues actions which are individually beneficial (R1 and R2). If the amount of activity grows too large for the system to support, **however, the "commons" becomes experiences diminishing benefits (B5 and B6).**

6



Escalation

Different parties take actions to counter a perceived threat

In the "**Escalation" archetype**, one party (A) takes actions that are perceived by the other as a threat. The other party (B) responds in a similar manner, increasing the threat to A and resulting in more threatening actions by A. The reinforcing loop is traced out by following the figure-8 outline produced by the two balancing loops.

3

Systems Thinking For The Internet and Future AI-Driven Self-Driving Networks

- *The Paradoxes of Internet Design*
- *Ethical and Policy Challenges in SDNs.*

On self-driving networks



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Why (and How) Networks Should Run Themselves

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Abstract

The proliferation of networked devices, systems, and applications that we depend on every day makes managing networks more important than ever. The increasing security, availability, and performance demands of these applications suggest that these increasingly difficult network management problems be solved in real time, across a complex web of interacting protocols and systems. Alas, just as the importance of network management has increased, the network has grown so complex that it is seemingly unmanageable. In this new era, network management requires a fundamentally new approach. Instead of optimizations based on closed-form analysis of individual protocols, network operators need data-driven, machine-learning-based models of end-to-end and application performance based on high-level policy goals and a holistic view of the underlying components. Instead of anomaly detection algorithms that operate on offline analysis of network traces, operators need classification and detection algorithms that can make real-time, closed-loop decisions. Networks should learn to drive themselves. This paper explores this concept, discussing how we might attain this ambitious goal by more closely coupling measurement with real-time control and by relying on learning for inference and prediction about a networked application or system, as opposed to closed-form analysis of individual protocols.

1 Introduction

Modern networked applications operate at a scale and scope we have never seen before. Virtual and augmented reality require real-time responsiveness, micro-services deployed using containers introduce rapid changes in traffic workloads, and the Internet of Things (IoT) significantly increases the number of connected devices while also raising new security and privacy concerns. The widespread integration of these applications into our daily lives raises the bar for network management, as users elevate their expectations for real-time interaction, high availability, resilience to attack, ubiquitous access, and scale. Network management has always been a worthwhile endeavor, but now it is mission critical.

Yet, network management has remained a Sisyphean task. Network operators develop and use scripts and tools to help them plan, troubleshoot, and secure their networks, as user demands and network complexity continue to grow. Networking researchers strive to improve the tuning, design, and measurement of network protocols, yet we continue to fall behind the curve, as the protocols, variable network conditions, and relationships between them and user quality of experience become increasingly complex. Twenty years ago, we had some hope of (and success in) creating clean, closed-form models of individual protocols, applications, and systems [4, 24]; today, many of these are too complicated for closed-form analysis. Prediction problems such as determining how search query response time would vary in response to the placement of a cache are much more suited to statistical inference and machine learning based on measurement data [29].

Of course, we must change the network to make network management easier. We have been saying this for years, as we continue to fall behind the curve. Part of the problem, we believe, is the continued focus on designing, understanding, and tweaking individual protocols—we focus on better models for BGP, optimizations for TCP, QUIC, DNS, or the protocols du jour. In fact, our troubles do not lie in the protocols. The inability to model holistic network systems, as opposed to individual protocols, has made it difficult for operators to understand what is happening in the network. Software-Defined Networking (SDN) helps by offering greater programmability and centralized control, yet controller applications still rely on collecting their own data and installing low-level match-action rules in switches and SDN does not change the fact that real networked systems are too complex to analyze with closed-form models.

As networking researchers, we must change our approach to these problems. An ambitious goal for network management is that of a *self-driving network*—one where (1) network measurement is task-driven and tightly integrated with the control of the network; and (2) network control relies on learning and large-scale data analytics of the entire networked system, as opposed to closed-form models of individual protocols. Recent initiatives have proffered this high-level goal [14, 28], drawing an analogy to self-driving cars, which can make decisions that manage uncertainty and mitigate risk to achieve some task (*e.g.*, transportation to some destination). This paper explores this goal in detail, developing the technical requirements for and properties of a self-driving network and outlining a broad, cross-disciplinary research agenda for the community that can move us closer to realizing this goal.

The networking research community has been developing the pieces of this puzzle for many years, from predictive models of application performance [19, 29] to statistical anomaly and intrusion detection algorithms based on analysis of network traffic [2, 7]. The state of the art, however, merely lays the foundation for the much more ambitious agenda of

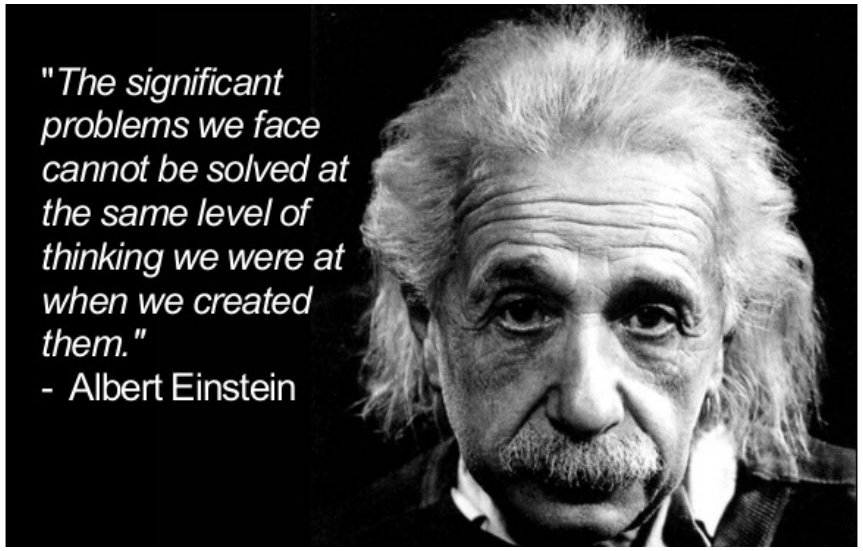
Abstract

The proliferation of networked devices, systems, and applications that we depend on every day makes managing networks more important than ever. The increasing security, availability, and performance demands of these applications suggest that these increasingly difficult network management problems be solved in real time, across a complex web of interacting protocols and systems. Alas, just as the importance of network management has increased, the network has grown so complex that it is seemingly unmanageable. In this new era, network management requires a fundamentally new approach. Instead of optimizations based on closed-form analysis of individual protocols, network operators need data-driven, machine-learning-based models of end-to-end and application performance based on high-level policy goals and a holistic view of the underlying components. Instead of anomaly detection algorithms that operate on offline analysis of network traces, operators need classification and detection algorithms that can make real-time, closed-loop decisions. Networks should learn to drive themselves. This paper explores this concept, discussing how we might attain this ambitious goal by more closely coupling measurement with real-time control and by relying on learning for inference and prediction about a networked application or system, as opposed to closed-form analysis of individual protocols.

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A systems approach to solving problems

	Conventional Thinking	***Systems Thinking***
Model of thinking	Linear, causal, open-looped, immediate feedback	Nonlinear, multi-causal, closed-looped with delayed feedback
Determining a problem's cause	Obvious and easy to trace	Indirect and non-obvious
Cause of problems	External to the system	Internal (System-as-a-cause thinking)
How to optimize?	By optimizing the parts	By optimizing relationships among the parts
Where to intervene?	Aggressive use of "obvious" solutions	Careful change applied at the "leverage points"
How to resolve problems?	Cure the symptoms	Fix the systemic causes



The paradoxes of Internet architecture



COLUMN: Standards

Paradoxes of Internet Architecture

Srinivasan Keshav
University of Waterloo

The architectural elements of the Internet that led to its great success are now, paradoxically, the source of its severest problems. For example, the use of autonomous systems to distribute governance has allowed rapid growth and scaling but has made the network unmanageable and unable to provide end-to-end quality of service. This article examines this and other key design elements of Internet architecture and shows how they have contributed to its success and how they now constrain it. I then use this framework to identify some key challenges that need to be addressed in the next phase of Internet research.

The paradoxes of Internet architecture

"Technology is both a burden and a blessing; not either-or, but this-and-that"—Neil Postman.

The top level goal for the DARPA Internet Architecture was to develop an effective technique for multiplexed utilization of existing interconnected networks.[3]

This goal is then further elaborated to encompass the following seven subgoals:

1. *Internet communication must continue despite loss of networks or gateways.*
2. *The Internet must support multiple types of communications service.*
3. *The Internet architecture must accommodate a variety of networks.*
4. *The Internet architecture must permit distributed management of its resources.*
5. *The Internet architecture must be cost effective.*
6. *The Internet architecture must permit host attachment with a low level of effort.*
7. *The resources used in the Internet architecture must be accountable.*

...

It is important to understand that these goals are in order of importance, and an entirely different network architecture would result if the order were changed.[3]

Three fundamental problems with the Internet today.

1. Spam.
2. Privacy and Security
3. Quality of Service

Keshav points out that paradoxically the Internet's architectural elements most responsible for its success are also responsible for its most vexing problems.

System archetypes in networking

Fixes That Backfire
A quick solution with unexpected long-term consequences

IPv4 NAT; Bufferbloat

Limits to Growth
Improvement accelerates and then suddenly stalls

IPv4

Escalation
Different parties take actions to counter a perceived threat

Adblocking & counterblocking

Shifting the Burden
Systems unconsciously favor short-term, addictive solutions

IPv4 NAT; cross-layer design;
Tussles in cyberspace

Tragedy of the Commons
Shared unmanaged resource collapses due to overconsumption

Spectrum commons;

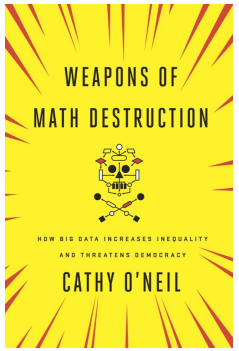
Success to the Successful
Things get better for "winners" and worse for "losers"

Network neutrality
Walled gardens

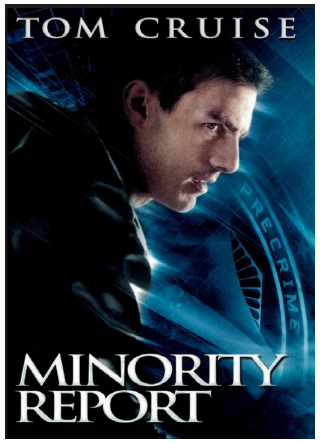
Ethical and policy challenges for SDNs

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THERE ARE ETHICAL CHOICES IN EVERY SINGLE ALGORITHM WE BUILD



The question of agency–i.e., **“who will take the ethical decision?”**– also looms large for self-driving networks.



In the movie Minority Report, the cop tackles and handcuffs individuals who have committed no crime (yet), proclaiming stuff like:

“By mandate of the District of Columbia Precrime Division, I’m placing you under arrest for the ***future*** *murder of Sarah Marks and Donald Dubin.”*

The arrested person confronts Cruise and asks:

“You ever get any false positives?”

AI can help strengthen stereotypes



Harvard Professor
Latanya Sweeney

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Arvind Narayanan ✓
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Following

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> Using voice and face recognition software, HireVue lets employers compare a candidate's word choice, tone, and facial movements with the body language and vocabularies of their best hires. The algorithm can analyze all of these candidates' responses and rank them, so that recruiters can spend more time looking at the top performing answers.

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Automated Inference on Criminality using Face Images

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(a) Three samples in criminal ID photo set S_c.

(b) Three samples in non-criminal ID photo set S_n

Figure 1. Sample ID photos in our data set.

Security challenges

"No problem stays solved in a dynamic environment."
—Russell Ackoff

Since algorithms are trained using historical datasets, self-driving networks are always vulnerable to future evolved adversarial attack.

Concluding remarks

Systems thinking helps us make sense of interdependency in complex system and the holistic behavior of a system by understanding the feedback loops at play.

With the rise of interest in self-driving networks, which will become part of the larger Internet, there is a need to rigorously look at how these technologies will affect—positively as well as negatively—all the stakeholders.

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Books

Business Dynamics: Systems Thinking and Modeling for a Complex World (Sterman),
Systems thinking made simple: New hope for solving wicked problems (Cabrera & Cabrera)

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